

TABLE I
Triboelectric Series

Most Positive (+)	
Air	+++
Human hands, skin	+++
Asbestos	+++
Rabbit fur	+++
Glass	+++
Human hair	+++
Mica	+++
Nylon	+
Wool	+
Lead	+
Cat fur	+
Silk	+
Aluminium	+
Paper	+
Cotton	+
Steel	-
Wood	-
Lucite	-
Sealing wax	-
Amber	-
Rubber balloon	-
Hard rubber	-
Mylar	-
Nickel	-
Copper	-
Silver	-
UV resist	-
Brass	-
Synthetic rubber	-
Gold, platinum	-
Sulphur	-
Acetate, rayon	-
Polyester	-
Celluloid	---
Polystyrene	---
Acrylic	---
Cellophane tape	---
Polyurethane	---
Polyethylene	---
Polypropylene	---
Polyvinylchloride (vinyl)	---
Silicon	---
Teflon	---
Silicone rubber	---
Most Negative (-)	

als involved to gain or lose electrons. Some materials have a greater tendency to gain electrons than the others. Similarly, some materials tend to lose electrons easier than the others.

The triboelectric series (Table I) lists various materials according to their tendency to gain or lose electrons. It usually lists materials in order of decreasing tendency to charge positively (lose electrons), and increasing tendency to

TABLE II
Common Charge Generation Situation and Corresponding Voltage Levels

Charge generation situations	10-25 per cent RH	65-90 per cent RH
Walking across carpet	35,000V	1500V
Walking across vinyl tile	12,000V	250V
Worker at bench	6000V	100V
Poly bag picked up from bench	20,000V	1200V

TABLE III
Common Sources of Static Charge

Object or process	Type
Work surfaces	Waxed, painted or plastic surfaces
Floors	Waxed, common vinyl tiles and sealed concrete
Clothes	Common clothes, non-conductive shoes and synthetic materials (for example, nylon)
Chairs	Vinyl, fibreglass and finished wood
Packaging	Common plastic bags, foam and trays
Assembly area	Spray cleaners, heat guns, blowers, plastic tools (for example solder suckers and brushes) and cathode ray tubes

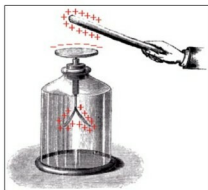


Fig. 2: Electrostatic induction

charge negatively (gain electrons).

Somewhere in the middle of the list are materials that do not show strong tendency to behave either way. Note that the tendency of a material to become positive or negative after triboelectric charging has nothing to do with the level of conductivity (or ability to discharge) of the material.

The charge created by triboelectric effect depends on various factors such as area of contact, the speed of separation, relative humidity and chemistry of the materials and surface work function. Table II shows the common charge generation situations and the resulting voltage levels generated due to static electricity.

The contribution of humidity in reducing charge accumulation is also known. It should be noted, however,

that some static generation occurs even at high relative humidity. It is not recommended to have relative humidity that is too low, say, below 30 per cent. ESD control becomes especially challenging at low-RH levels. A relative humidity between 40 per cent and 60 per cent is recommended for a typical assembly area.

When an electrostatically-charged person or object touches an ESD-sensitive device, there is a possibility of the electrostatic charge being drained through sensitive circuitry in the device. If the ESD possesses sufficient energy, damage could occur to the device due to localised overheating. Generally, devices with finer geometries are more susceptible to damage from ESD. The common sources of static charge in a lab are mentioned in Table III.

Electrostatic induction. This is another phenomenon responsible for static charge build up. It occurs when an electrically charged object comes near a conductive object isolated from the ground. The charged object produces an electrostatic field that causes electrical charges on the surface of the other object to redistribute as shown in Fig. 2. As such, the net charge on the object will be zero, but it will have surfaces with excess positive and negative charge and so an ESD event can happen from those surfaces.